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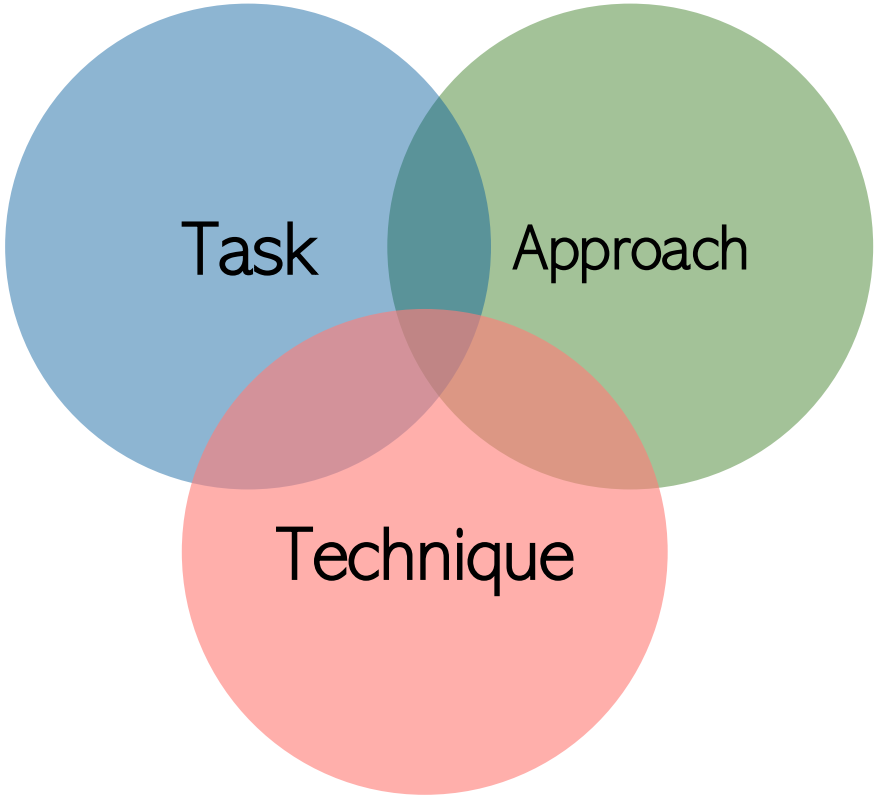


https://www.youtube.com/watch?v=JBdR9Id0Glg
Samuel Watson: 20 years old vs. Reza Alipour: 32 years old

4

Outline: W07C: New Technique: Logistic Regression (LR)

Classification
Regression
Clustering
Ranking (Recommendation)
Vector Representation and Dimensionality Reduction
Language Modeling



Full/Self/Semi-**Supervised Learning**
Unsupervised Learning
Reinforcement Learning

k-Nearest Neighbor, Naive Bayes, **Logistic Regression**
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means

Outline: W07C

I) Task: Classification

II) Supervision: Fully Supervised

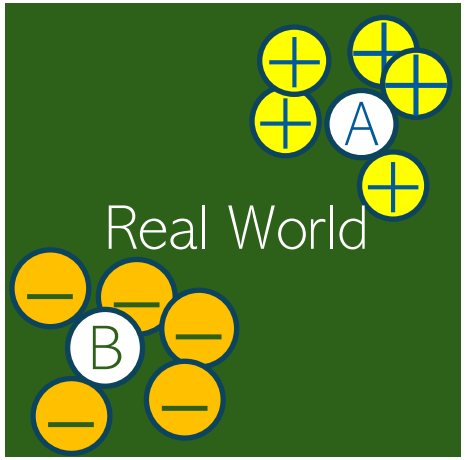
III) Technique: Logistic Regression → Neural Networks

IIII) Evaluation

Metrics (Few New Metrics), Methodology

kNN is a Locally Affinitive Learner

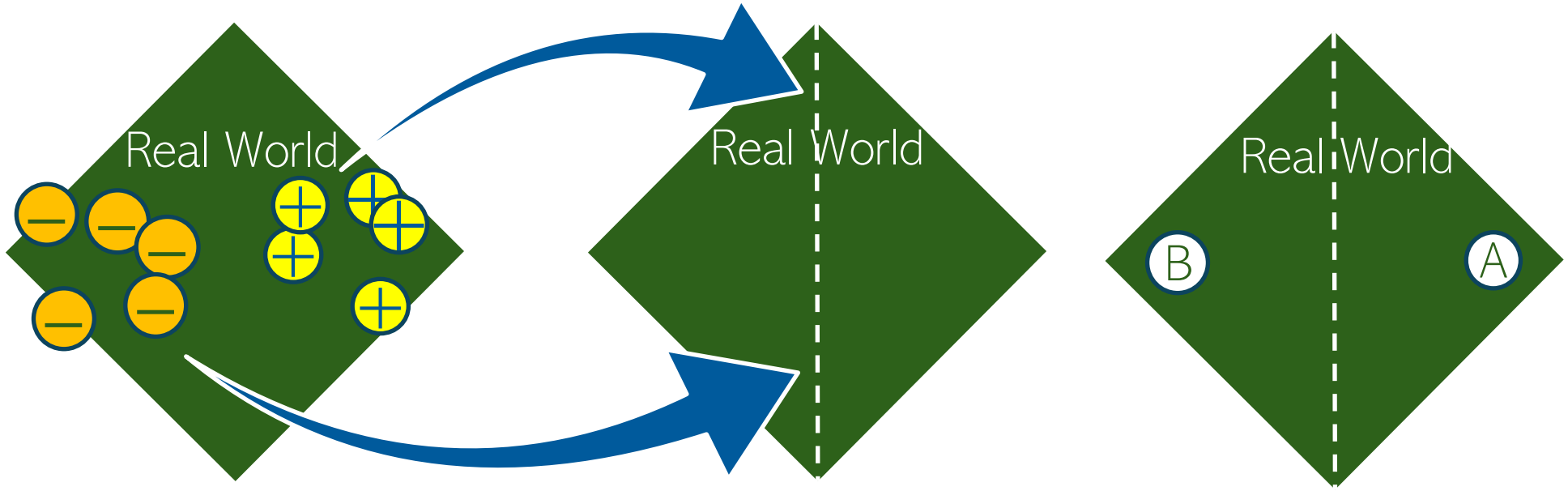
2-NN(A) = 1
2-NN(B) = 0



7

A Technique based on a Global Discriminator

Summarizing the Entire World into a Discrimination Boundary



f(A) = Far Right = Very Pos
f(B) = Far Left = Very Neg

Discriminator in Euclidian Space

1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0$

The universe can be summarized (learned) into a number:

- 1 parameter (memory cell, weight)

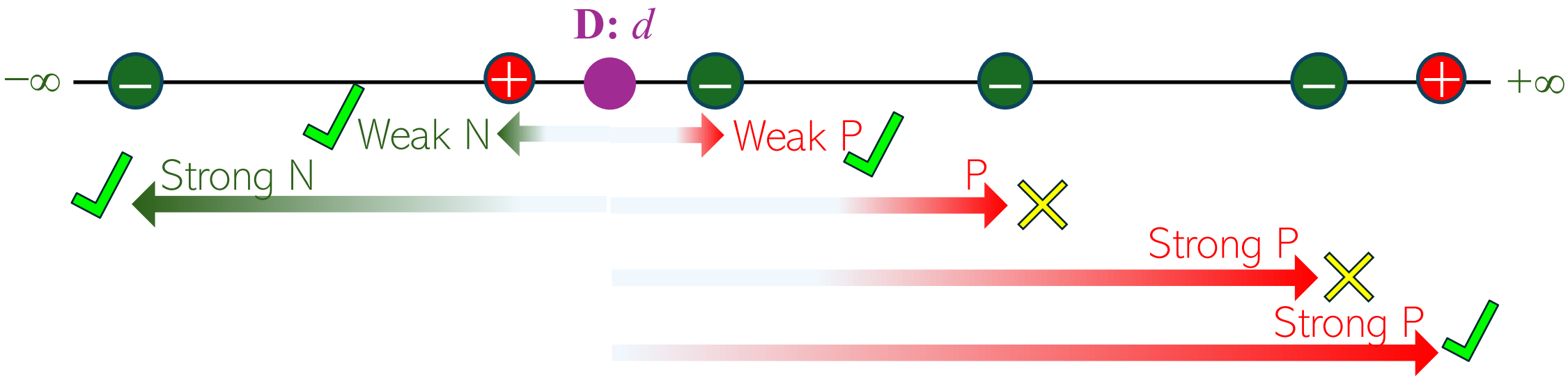


$$f(\mathbf{o}) = d(\mathbf{o}, \mathbf{D})$$
$$= o_i - d$$
$$= \begin{cases} > 0\text{: larger positive, stronger discrimination for being positive} \\ < 0\text{: smaller negative, stronger discrimination for being negative} \end{cases}$$



Discriminator in Euclidian Space

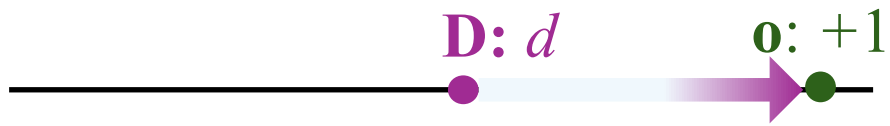
1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0$

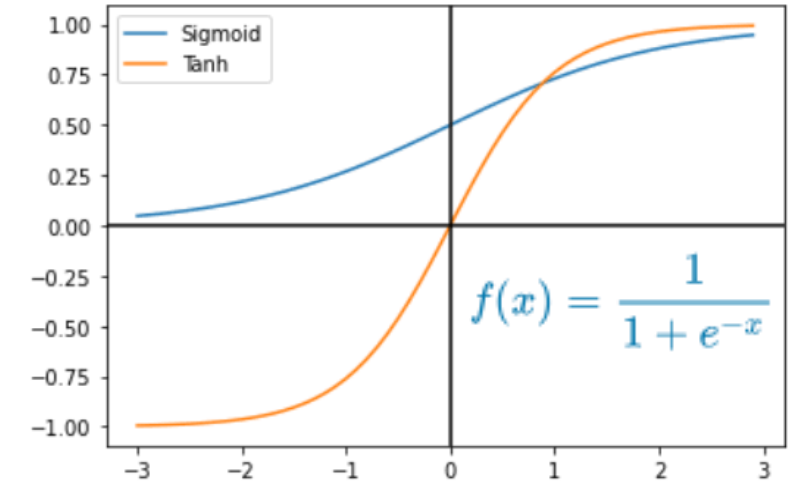


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Discriminator in Euclidian Space

1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0 \rightarrow \mathbb{R}^{[0,+1]}$



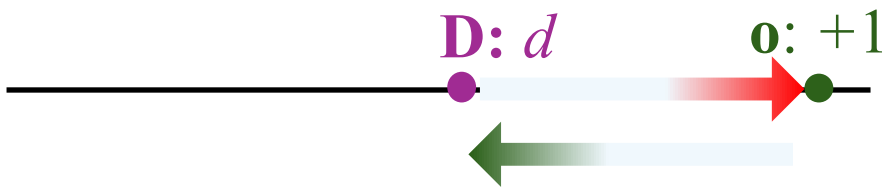


$$
\begin{aligned}
f(\mathbf{o}) &= d(\mathbf{o}, \mathrm{D}) \\
&= \textit{logistic curve}(o_i - d) \\
&= \textit{logistic regression}(o_i - d) \\
&= \textit{sigmoid}(o_i - d) \\
&= \sigma\,(o_i - d) \\
&= \begin{cases} \text{The more toward } +1\text{, stronger positive} \\ \text{The less toward } +1\text{, weaker positive} \end{cases}
\end{aligned}
$$

Discriminator in Euclidian Space

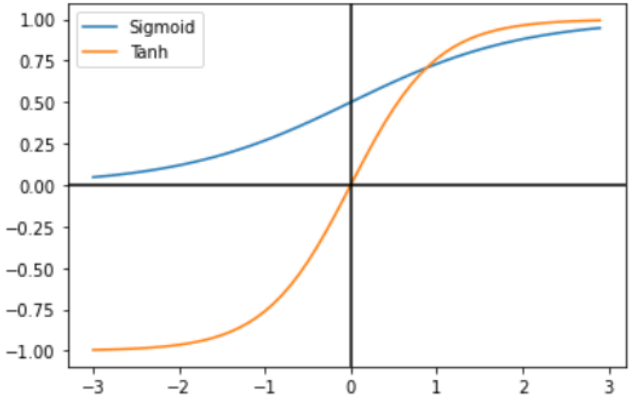
1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability

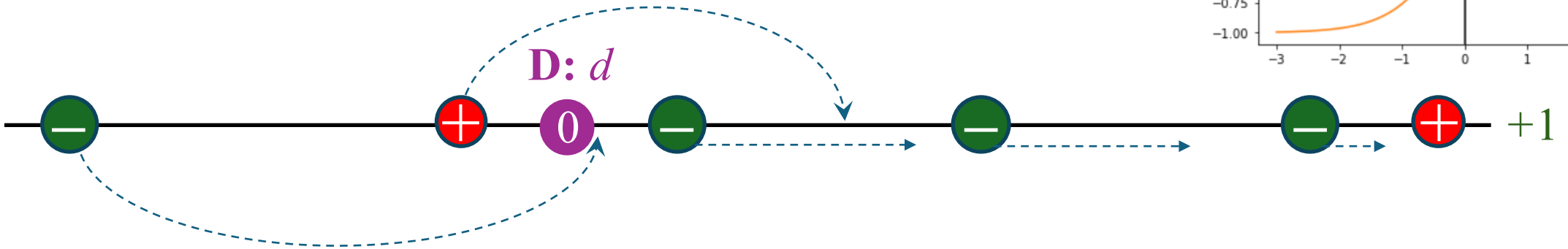
$$
\begin{aligned}
f(\mathbf{o}) &= d(\mathbf{o}, \mathrm{D}) \\
&= \mathit{logistic\ curve}(o_i - d) \\
&= \mathit{logistic\ regression}(o_i - d) \\
&= \mathit{sigmoid}(o_i - d) \\
&= \sigma\,(o_i - d) \\
&= \begin{cases} \text{The more toward } +1\text{, stronger probability for being positive} \\ \underbrace{1 - \text{probability for being positive}}_{\text{Being negative}} = 1 - \mathit{sigmoid}(o_i - d). \end{cases}
\end{aligned}
$$

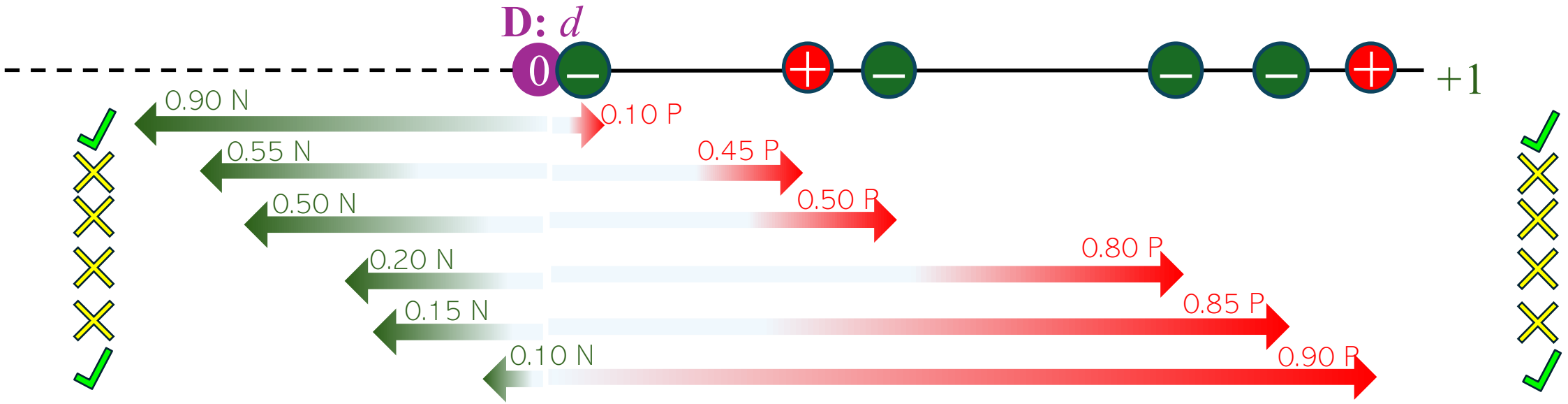


Discriminator in Euclidian Space

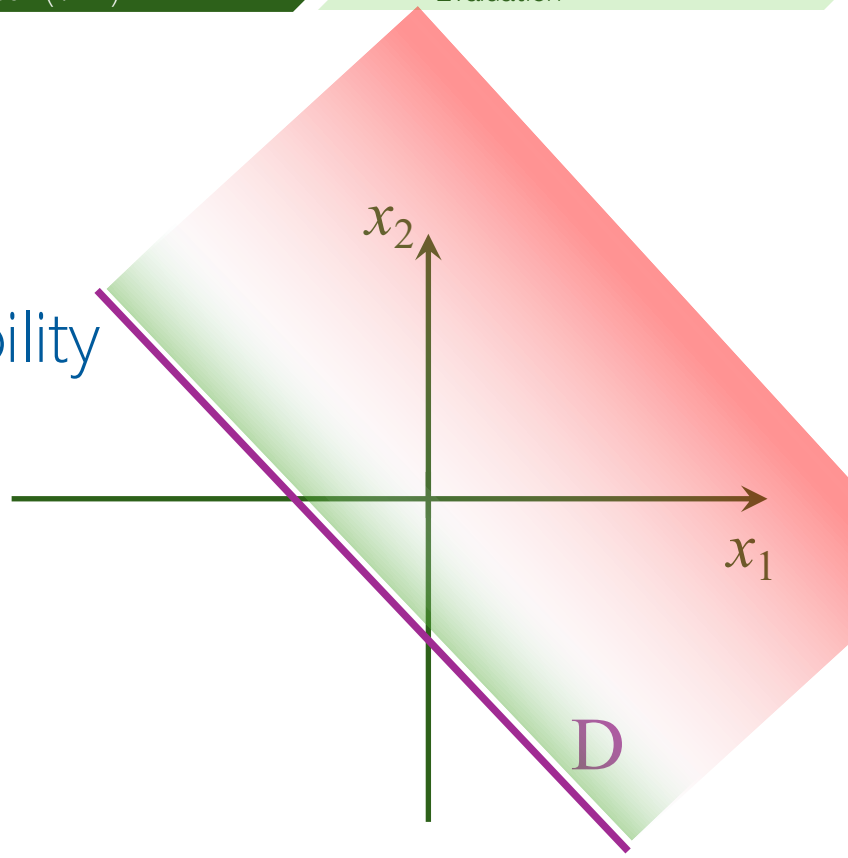
1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability







Discriminator: Euclidian Space



2D Euclidian Distance $\mathbb{R}^2 \rightarrow \mathbb{R}^1 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability

$$f(\mathbf{o}) = d(\mathbf{o}, \mathrm{D})$$

$$\approx sigmoid([x_1 \quad x_2 \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix})$$

$$\approx \sigma([x_1 \quad x_2 \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix})$$

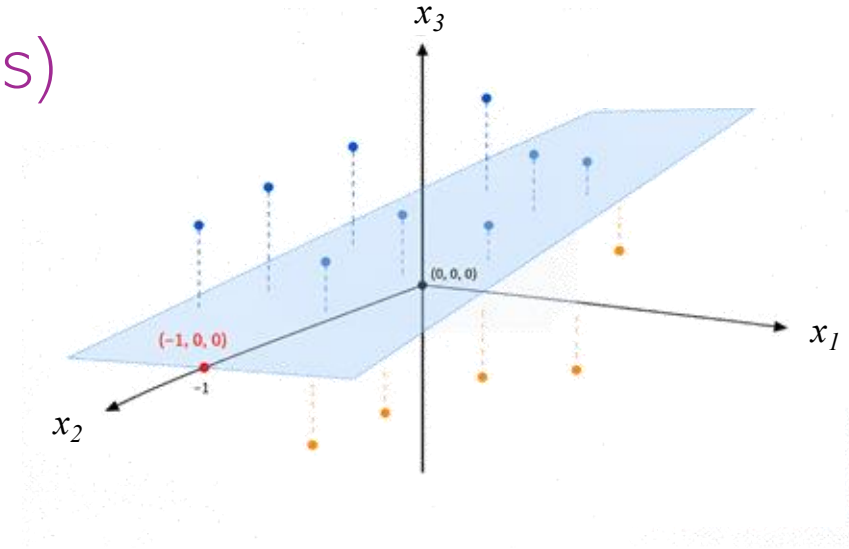
$$\approx \begin{cases} \text{The more toward } +1\text{, stronger } \textbf{probability} \text{ for being positive} \\ \underbrace{1 - \text{probability for being positive}}_{\text{Being negative}} = 1 - sigmoid([x_1 \quad x_2 \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}) \end{cases}$$

Discriminator: Euclidian Space

2D Euclidian Distance $\mathbb{R}^3 \longrightarrow \mathbb{R}^2$

The universe can be summarized (learned) into a plane:

- 4 parameters (coefficients, weights, memory cells)

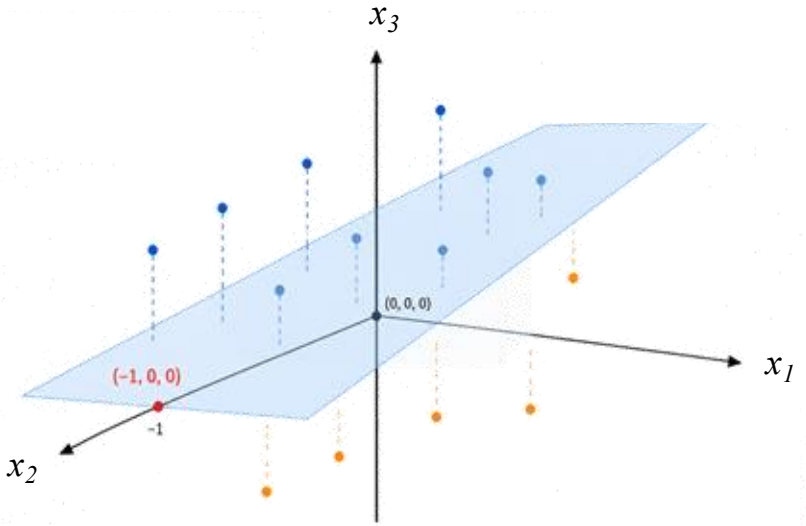


$\longrightarrow \mathbb{R}^{[0,+1]} \longrightarrow$ Probability

$$f(\mathbf{o}) = d(\mathbf{o}, \mathbf{D}) \approx \sigma([x_1 \quad x_2 \quad x_3 \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix})$$

$$\approx \begin{cases} \text{The more toward } +1\text{, stronger probability for being positive} \\ \underbrace{1 - \text{probability for being positive}}_{\text{Being negative}} = 1 - sigmoid(\ldots) \end{cases}$$

Discriminator: Euclidian Space



What is, or who knows, the true discriminator **D**?

- Point, what is that point $d=w_1$?
- Line, what is that line w_1, w_2, w_3?
- Plane, what is that plane w_1, w_2, w_3, w_4?
- Hyperplane, what is that hyperplane w_1, w_2, …, w_{n+1}?

No one!

Aletheia, the goddess of truth
God, in the Abrahamic religions as the all-powerful, all-knowing, eternal deity, and single creator of the universe
Omniscient /äm'niSHənt/: omni=all, scientia= knowledge or to know
The Oracle of Delphi
Teacher
Supervisor
Human (Expert) Annotator
Majority Vote on Human (Expert) Annotators
LLM-as-a-Judge



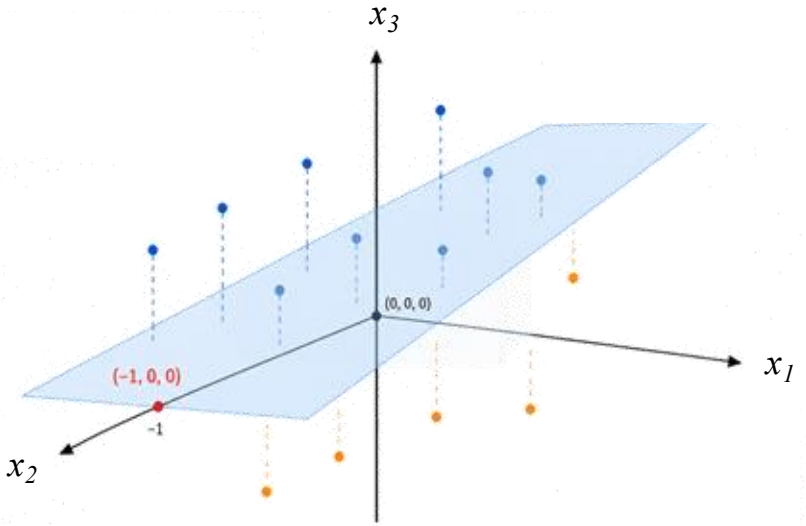


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Discriminator: Euclidian Space

What is, or who knows, the true discriminator **D**?

- Point, what is that point $d=w_1$?
- Line, what is that line w_1, w_2, w_3?
- Plane, what is that plane w_1, w_2, w_3, w_4?
- Hyperplane, what is that hyperplane w_1, w_2, …, w_{n+1}?



We find it through an iterative process:

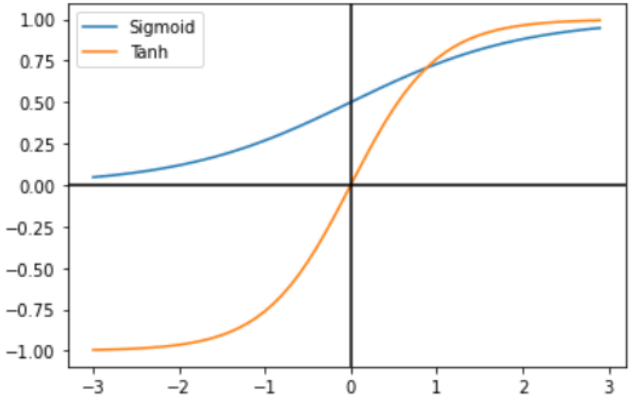
1. Initial random guess
2. Evaluate on the train set
3. Tune the discriminator's parameters toward *almost* overfitting
4. Back to (2)

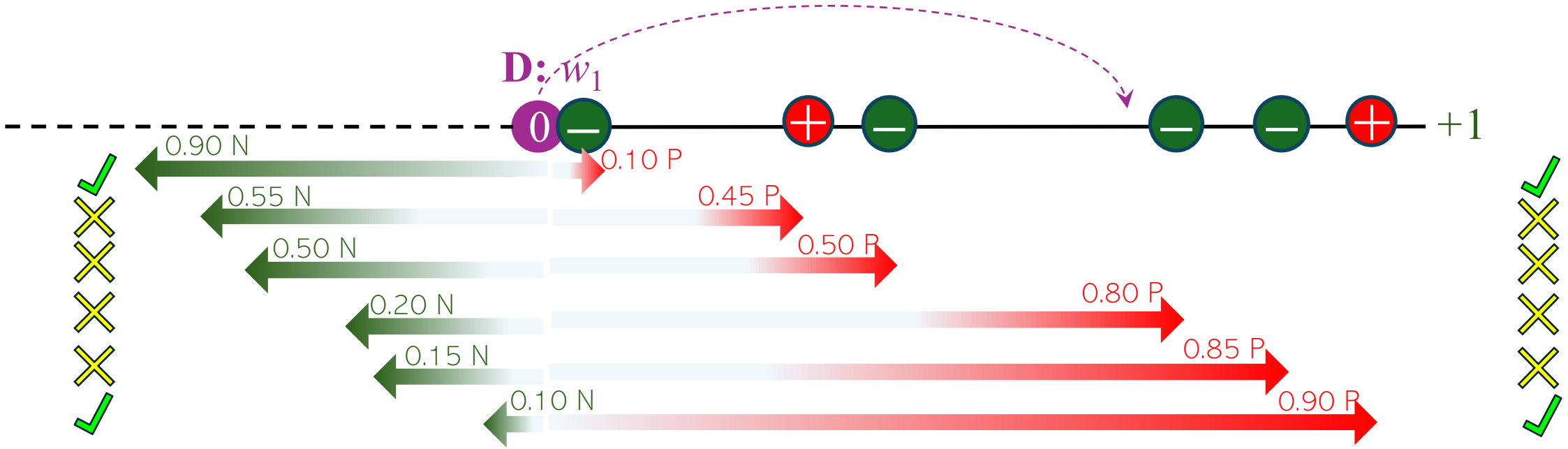




Discriminator in Euclidian Space

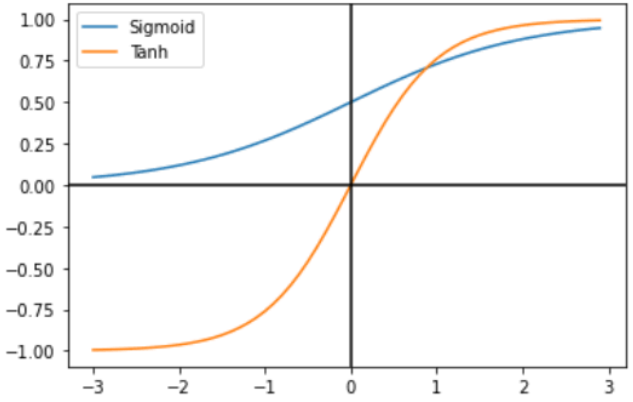
1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability

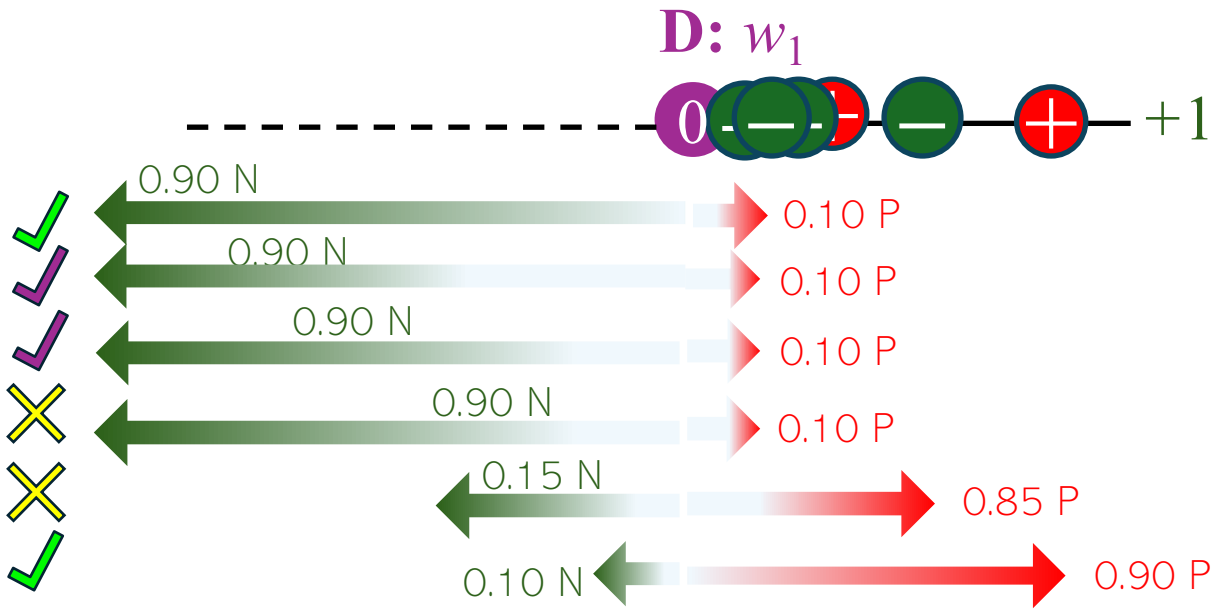




Discriminator in Euclidian Space

1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability







Outline: W08A

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: Logistic Regression

Learning the Discriminator from Training Data (from Experience)

IIII) Evaluation

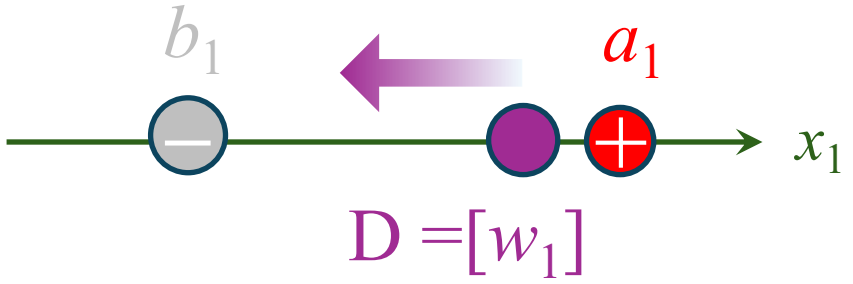
Metrics (Few New Metrics), Methodology

20

Learning the Discriminator

Focus on catching positive observations:

- Maximizing number of true positives
- Minimizing number of false negatives



A. Error = $[f^{+}(\boldsymbol{o}^{+}) - \text{'Positive'}]$

$= [f^{+}(\boldsymbol{o}^{+}) - 1.0]$ but this is negative, let's make it positive number

$= [1.0 - f^{+}(\boldsymbol{o}^{+})]$

$= [1.0 - \sigma(a_1 - w_1)]$

$= e(w_1) \geq 0$

B. Modify w_1 such that $e(w_1) \rightarrow 0.0$

C. Back to A

21

Learning the Discriminator

Focus on catching positive observations:

× May result in pessimistic worldview



$$D = [w_1 \rightarrow -\infty]$$

A. Error $= [f^+(\boldsymbol{o}^+) - \text{'Positive'}]$

$= [f^+(\boldsymbol{o}^+) - 1.0]$ but this is negative, let's make it positive number

$= [1.0 - f^+(\boldsymbol{o}^+)\ ]$

$= [1.0 - \sigma\,(a_1 - w_1)]$

$= e\,(w_1) \geq 0$

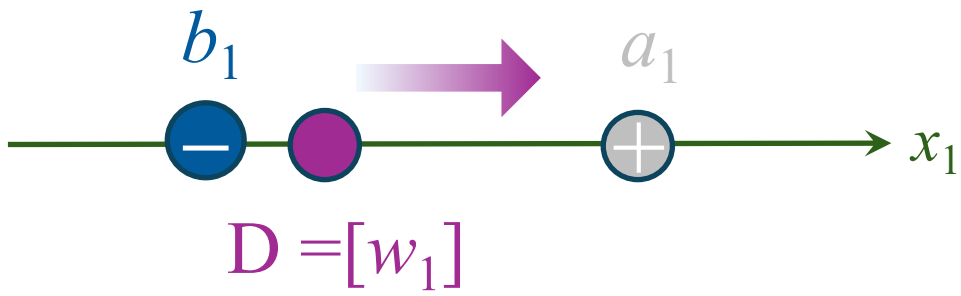
B. Modify w_1 such that $e\,(w_1) \rightarrow 0.0$ results in $w_1 \rightarrow -\infty$

C. Back to A

Learning the Discriminator

Focus on catching negative observations:

- Maximizing number of true negatives
- Minimizing number of false positives



A. Error = $[f^{+}(\boldsymbol{o}^{-}) - \text{'Negative'}]$

$= [f^{+}(\boldsymbol{o}^{-}) - 0.0]$ always positive number

$= [\sigma\,(b_1 - w_1)]$

$= e\,(w_1) \geq 0$

B. Modify w_1 such that $e\,(w_1) \rightarrow 0.0$

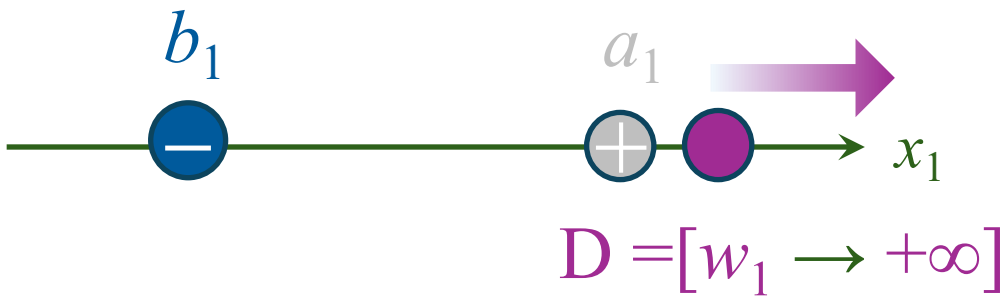
C. Back to A

23

Learning the Discriminator

Focus on catching negative observations:

× May result in optimistic worldview



A. Error $= [f^{+}(\boldsymbol{o}^{-}) - \text{'Negative'}]$

$= [f^{+}(\boldsymbol{o}^{-}) - 0.0]$ always positive number

$= [\sigma\,(b_1 - w_1)]$

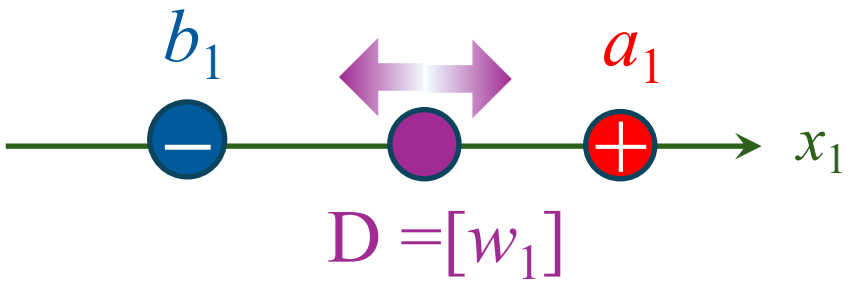
$= e\,(w_1) \geq 0$

B. Modify w_1 such that $e\,(w_1) \rightarrow 0.0$ results in $w_1 \rightarrow +\infty$

C. Back to A

Learning the Discriminator

Both positive and negative observations



A. Error $= [f^{+}(\boldsymbol{o}^{+}) - \text{'Positive'}] + [f^{+}(\boldsymbol{o}^{-}) - \text{'Negative'}]$

$= [1.0 - f^{+}(\boldsymbol{o}^{+})] + [f^{+}(\boldsymbol{o}^{-}) - 0.0]$

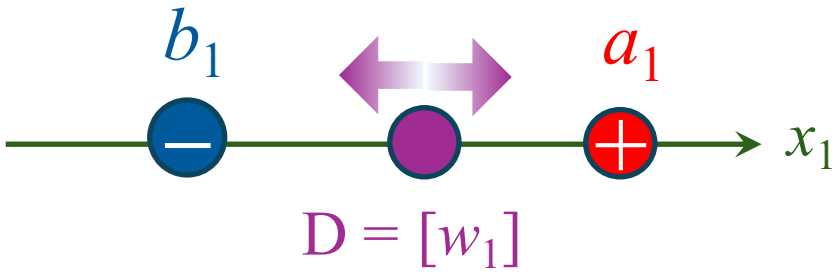
$= [1.0 - \sigma(a_1 - w_1)] + [\sigma(b_1 - w_1)]$

$= e(w_1) \geq 0$

B. Modify w_1 such that $e(w_1) \rightarrow 0.0$

C. Go back to A

Learning the Discriminator → An Optimization Problem
Data Mining Task → Mathematic Task



$$e(w_1) = [1.0 - f^+(\boldsymbol{o}^+)] + [f^+(\boldsymbol{o}^-) - 0.0] = 0.0$$
$$= [1.0 - f^+(\boldsymbol{o}^+)] + [f^+(b_1) - 0.0] = 0.0$$
$$= [1.0 - \sigma(a_1 - w_1)] + [\sigma(b_1 - w_1)] = 0.0$$

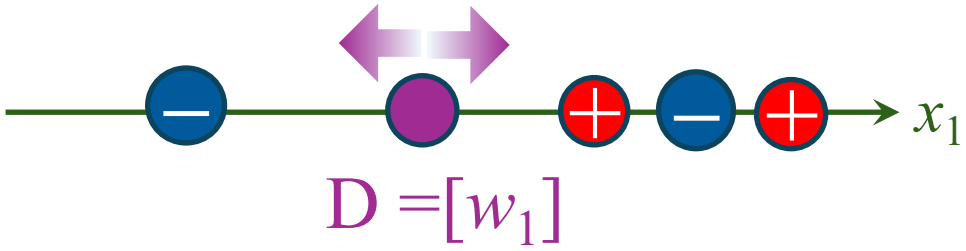
$e(w_1) > 0$ and never becomes 0. Indeed, $\forall w_1 \in \mathbb{R}$ such that $e(w_1) = 0$

The intuitive guess would be $w_1 = \frac{a_1 + b_1}{2}$, just in the middle of a_1 and b_1? Try it for any random pair of numbers and see the error can be zero or at least close to zero …

Learning the Discriminator → An Optimization Problem
Data Mining Task → Mathematic Task

$$e(w_1) = \sum_{o_1 \in \mathbf{O}^+} [1.0 - f^+(\boldsymbol{o}^+)] + \sum_{o_1 \in \mathbf{O}^-} [f^+(\boldsymbol{o}^-) - 0.0] = 0.0$$

$$= \sum_{o_1 \in \mathbf{O}^+} [1.0 - \sigma(o_1 - w_1)] + \sum_{o_1 \in \mathbf{O}^-} [\sigma(o_1 - w_1) - 0.0] = 0.0$$



No solution may exist (no real value of w_1 satisfies the equation exactly.)

No closed-form solution may exist (a solution exists, but difficult to obtain.)



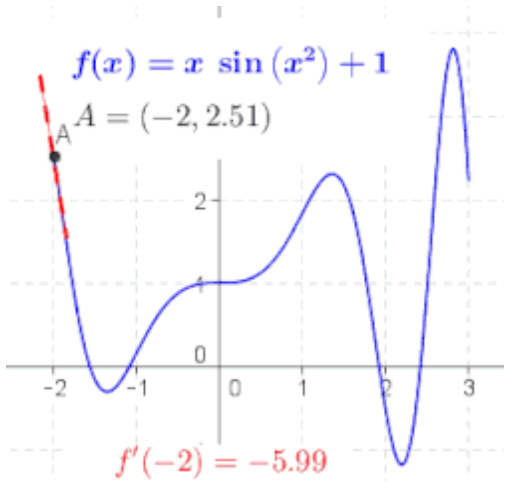
Lucy Smith and Pauline Rankin rock climbing in skirts on Salisbury Crags, Edinburgh. 1908.

An Optimization Problem

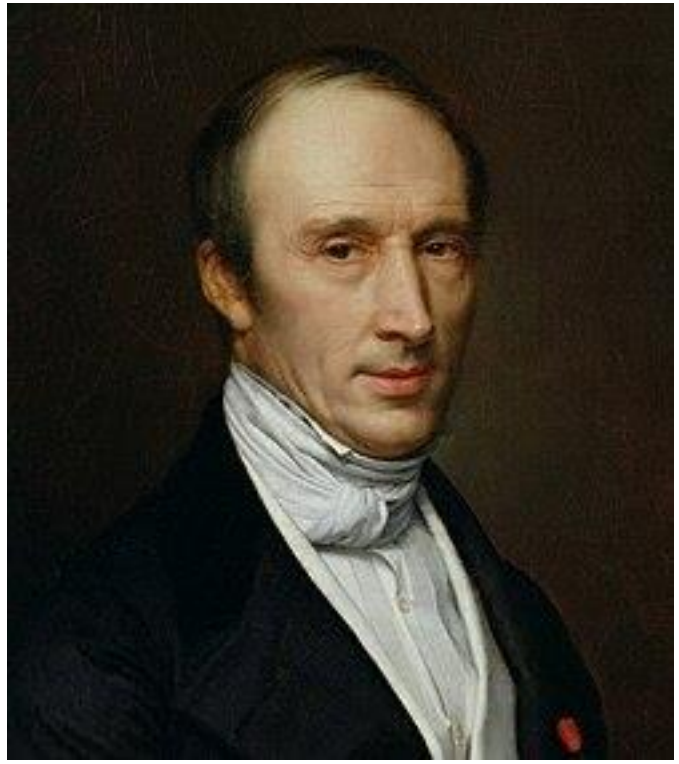


Pierre de Fermat (1700 – 1665)
French magistrate, polymath, and above all, mathematician also a poet
He is notably credited with Fermat's principle in optics.
He is especially known for having stated Fermat's Last Theorem, the proof was only established more than 300 years later.

Fermat noticed that at the very peak (maximum) or very trough /traaf/ (minimum) of a curve, the curve briefly flattens out. If you shift your position by an infinitely small amount, the output value barely changes



Baron Augustin-Louis Cauchy (1789 – 1857)
French mathematician, engineer, and physicist.
Proved the key theorems of calculus more than any other mathematician.
Approximately **800 research articles and 5 complete textbooks**.



While Fermat mastered finding minimums for single-variable equations, optimizing complex functions with multiple variables required the gradient. Cauchy invented the **Gradient Descent algorithm** to solve complex astronomical and algebraic equations, recognizing that moving in the **opposite direction of the gradient vector** provides the **steepest route** to a minimum point.



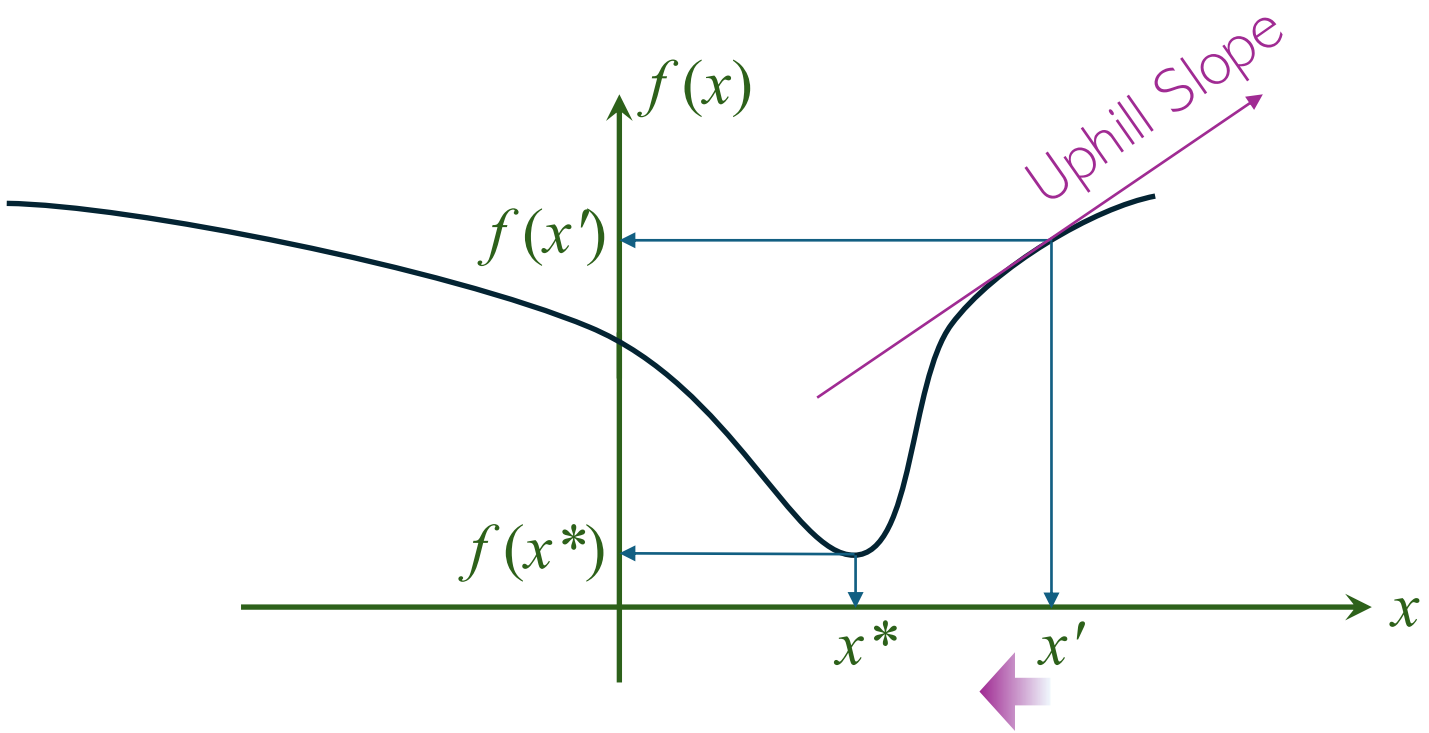
Mathematic Task: Finding the Minimum of a Function

Differentiation, Derivative

Where $f(x)$ become minimum value?

At what $x=x^*$, $f(x=x^*)$ become minimum value?

If at x', how to modify $x' \rightarrow x^*$ such that $f(x)$ get closer to minimum?



$$= lim_{\Delta x \to 0} \frac{f(x'+\Delta x) - f(x')}{x' - (x'+\Delta x)}$$

$$= lim_{\Delta x \to 0} \frac{f(x'+\Delta x) - f(x')}{\Delta x}$$

$$= \frac{d\,f(x')}{d\,x} > 0$$

One step down:

$$x' \rightarrow x' - \frac{d\,f(x')}{d\,x}$$

